

Catrina McKenzie Hacker

SUMMARY

Electrophysiologist and computational neuroscientist working at the intersection of visual and cognitive neuroscience to study how neural populations dynamically and flexibly support cognitive processes like visual memory. 7+ years of experience thinking critically about complex datasets, conducting electrophysiological and behavioral experiments, and communicating complicated ideas. Passionate about translation across model species, collaborative science, and clear scientific communication.

EDUCATION

Ph.D. in Neuroscience, *University of Pennsylvania* **2019 – 2026**

Thesis advisor: Nicole C. Rust, Ph.D.

Thesis title: *The Neural Correlates of Visual Recognition Memory as a Means of Bridging Observations Across Primate Species*

B.S. in Neuroscience w/honors, *University of Southern California*, Summa cum laude **2015 – 2019**

Honors thesis advisor: Irving Biederman, Ph.D.

Undergraduate research advisor: Sarah W. Bottjer, Ph.D.

RESEARCH POSITIONS

Postdoctoral Scholar, *University of Chicago*

2026 – Present

PI: Marlene Cohen, Ph.D.

Research focus: How does the brain support visual memories that are both highly specific and easily generalizable? What role do neurotransmitters play in mediating this balance and how does that relate to memory changes observed with aging?

PUBLICATIONS

Hacker, C.M., Bohn, S., Foster, B.L. & Rust, N.C. (2026). Neural representations of visual memory in inferotemporal cortex reveal a generalizable framework for translation between spikes and field potentials. *bioRxiv*. doi: [10.64898/2026.01.03.697516](https://doi.org/10.64898/2026.01.03.697516).

Jannuzi, B.G.L., **Hacker, C.M.**, Bohn, S., Meyer, T., Hay, M.L. & Rust, N.C. (2025). Sharpened visual memory representations are reflected in inferotemporal cortex. *Journal of Neuroscience*. doi: [10.1523/JNEUROSCI.0833-25.2025](https://doi.org/10.1523/JNEUROSCI.0833-25.2025). Chosen for a [spotlight](#).

Bohn, S., **Hacker, C.M.**, Jannuzi, B.G.L., Meyer, T., Hay, M.L. & Rust, N.C. (2025). Resolving a paradox about how vision is transformed into familiarity. *bioRxiv*. doi: [10.1101/2025.06.13.659490](https://doi.org/10.1101/2025.06.13.659490).

Rust, N.C., Yang, Y., **Hacker, C.M.** & Stuphorn, V. (2025). The representation of mood in primate anterior insular cortex. *bioRxiv*. doi: [10.1101/2025.04.22.650010](https://doi.org/10.1101/2025.04.22.650010).

Hacker, C.M. & Rust, N.C. (2022). Ritalin as a causal perturbation. *Trends in Cognitive Sciences*, Research Spotlight. doi: [10.1016/j.tics.2022.04.002](https://doi.org/10.1016/j.tics.2022.04.002).

Hacker, C.M., Biederman, I., Zhu, T., Nelken, M. & Meschke, E.X. (2022). The sizable difficulty in matching unfamiliar faces differing only moderately in orientation in depth is a function of image dissimilarity. *Vision Research*. doi: [10.1016/j.visres.2021.09.005](https://doi.org/10.1016/j.visres.2021.09.005).

PUBLICATIONS (CONT.)

- Hacker, C.M.**, Meschke, E.X. & Biederman, I. (2019). A Face in a (Temporal) Crowd. *Vision Research*, doi: [10.1016/j.visres.2018.02.007](https://doi.org/10.1016/j.visres.2018.02.007).
- Hacker, C.M.** & Biederman, I. (2019). The proficiency for distinguishing faces is independent of the proficiency for remembering them. *PysArxiv*. doi: [10.31234/osf.io/9bwct](https://doi.org/10.31234/osf.io/9bwct).
- Hacker, C.M.** & Biederman, I. (2019). The invariance of recognition to the stretching of faces is not explained by familiarity or warping to an average face. *PsyArxiv*. doi: [10.31234/osf.io/e5hgx](https://doi.org/10.31234/osf.io/e5hgx).
- Biederman, I., Shilowich, B.E., Herald, S.B., Margalit, E., Maarek, R., Meschke, E.X. & **Hacker, C.M.** (2018). The Cognitive Neuroscience of Person Identification. *Neuropsychologia*, 116, 205-214. doi: [10.1016/j.neuropsychologia.2018.01.036](https://doi.org/10.1016/j.neuropsychologia.2018.01.036).

POSTERS AND PRESENTATIONS

Talks

- Hacker, C.M.**, Bohn, S., Foster, B.L. & Rust, N.C. (2025). Field potentials sensitively and efficiently capture visual memory representations in inferotemporal cortex. Talk presented at the annual meeting of the Society for Neuroscience, San Diego, CA. November. Co-organized the nanosymposium session, "Neuronal dynamics underlying memory".
- Hacker, C.M.**, Jannuzi, B.G.L., Meyer, T., Hay, M.L. & Rust, N.C. (2024). Identifying the neural correlates of contextual influences on image memorability. Invited talk at the Workshop on Synergizing the Human Brain and Artificial Neural Networks, virtual.
- Hacker, C.M.**, Jannuzi, B.G.L., Meyer, T., Hay, M.L. & Rust, N.C. (2023). Identifying the neural correlates of contextual influences on image memorability. Talk presented at the annual Neuroscience/Vision/Auditory Training Grant Retreat, Philadelphia, PA. May.
- Hacker, C.M.**, Jannuzi, B.G.L., Meyer, T., Hay, M.L. & Rust, N.C. (2022). Worse remembering of a dog when viewed in a sequence of dogs is dominated by changes in memory mechanisms as opposed to sensory adaptation. Talk presented at the annual meeting of the Vision Sciences Society, St. Petersburg Beach, FL. May. doi: [10.1167/jov.22.14.4075](https://doi.org/10.1167/jov.22.14.4075).

Posters Presented

- Hacker, C.M.**, Bohn, S., Foster, B.L., Rust, N.C. (2025). A systematic comparison of the visual memory representations in spikes and LFPs. Poster presented at the meeting of the Simian Collective. Durham, NC. August.
- Hacker, C.M.**, Bohn, S., Foster, B.L., Rust, N.C. (2025). Would the same inferences about visual memory have been made with LFPs as compared to spikes? Poster presented at the 8th Annual Conference on Cognitive Computational Neuroscience, Amsterdam, The Netherlands. August.
- Hacker, C.M.**, Bohn, S., Jannuzi, B.G.L., Meyer, T., Hay, M.L. & Rust, N.C. (2024). A systematic comparison of the visual memory information in spikes, high gamma, and LFPs in inferotemporal cortex. Poster presented at the annual meeting of the Society for Neuroscience, Chicago, IL. October.
- Hacker, C.M.**, Jannuzi, B.G.L., Meyer, T., Hay, M.L. & Rust, N.C. (2023). A role for cortical pattern separation in enhancing visual memory. Poster presented at 150th Scheie Eye Anniversary Meeting, Philadelphia, PA. May.

POSTERS AND PRESENTATIONS

Posters Presented (cont.)

- Hacker, C.M.**, Jannuzi, B.G.L., Meyer, T., Hay, M.L. & Rust, N.C. (2023). A role for cortical pattern separation in enhancing visual memory. Poster presented at COSYNE, Montreal, Québec, Canada. March.
- Hacker, C.M.**, Jannuzi, B.G.L., Meyer, T., Hay, M.L. & Rust, N.C. (2022). Evidence that the extrinsic effects on memorability are computed in inferotemporal cortex and inherited by the hippocampus. Poster presented at the annual meeting of the Society for Neuroscience, San Diego, CA. November.
- Hacker, C.M.** & Biederman, I. (2019). The capacity for face perception is independent of the capacity for face memory. Poster presented at the annual meeting of the Vision Sciences Society, St. Petersburg Beach, FL. May. doi: [10.1167/19.10.139a](https://doi.org/10.1167/19.10.139a).
- Hacker, C.M.**, Meschke, E.X. & Biederman, I. (2018). Recognition of Stretched Faces. Poster presented at the annual meeting of the Vision Sciences Society, St. Petersburg Beach, FL. May. doi: [10.1167/18.10.160](https://doi.org/10.1167/18.10.160).
- Meschke, E.X.*, **Hacker, C.M.***, Juarez, J.J., Maarek, R.S. & Biederman, I. (2017). Detecting Unspecified Familiar Faces. Poster presented at the annual meeting of the Vision Sciences Society, St. Petersburg Beach, FL. May. doi: [10.1167/17.10.1027](https://doi.org/10.1167/17.10.1027).

Posters Co-Authored

- Rust, N.C., Yang, Y., **Hacker, C.M.** & Stuphorn, V. (2026). The representation of mood in primate anterior insular cortex. Selected for a short talk at The Neurobiology of Mental Health Lake conference, Thun, Switzerland. May.
- Bohn, S., **Hacker, C.M.**, Jannuzi, B.G.L., Meyer, T., Hay, M.L. & Rust, N.C. (2025). Resolving a paradox about how vision is transformed into recognition memory. Poster presented at the annual meeting of the Society for Neuroscience, San Diego, CA. November.
- Bohn, S., **Hacker, C.M.**, Jannuzi, B.G.L., Meyer, T., Hay, M.L. & Rust, N.C. (2022) Disambiguating familiarity from visual modulation: A role for the hippocampus in recognition memory. Poster presented at the annual meeting of the Society for Neuroscience, San Diego, CA. November.
- Jannuzi, B.G.L., **Hacker, C.M.**, Meyer, T., Hay, M.L. & Rust, N.C. (2022) Neural analogs of memory sharpening behavior emerge earlier in inferotemporal cortex than the hippocampus. Poster presented at the annual meeting of the Vision Sciences Society, St. Petersburg Beach, FL. May. doi: [10.1167/jov.22.14.4025](https://doi.org/10.1167/jov.22.14.4025).
- Jannuzi, B.G.L., Meyer, T., Hay, M.L., **Hacker, C.M.** & Rust, N.C. (2021) The remarkable visual specificity of visual recognition memory behavior is shaped by representational sharpening, reflected in inferotemporal cortex. Poster presented at the annual meeting of the Society for Neuroscience, virtual. November.
- Biederman, I., Zhu, T., Nelken, M., Meschke, E.X. & **Hacker, C.M.** (2019). The cost of matching depth-rotated faces: A simple, additive function of image similarity. Poster presented at the annual meeting of the Vision Sciences Society, St. Petersburg Beach, FL. May. doi: [10.1167/19.10.136b](https://doi.org/10.1167/19.10.136b).
- Meschke, E.X., **Hacker, C.M.** & Biederman, I. (2018). How Many Faces Can We Recognize? Poster presented at the annual meeting of the Vision Sciences Society, St. Petersburg Beach, FL. May. doi: [10.1167/18.10/158](https://doi.org/10.1167/18.10/158).

POSTERS AND PRESENTATIONS

Posters Co-Authored (cont.)

- Zhu, T., Nelken, M., **Hacker, C.M.**, Meschke, E.X. & Biederman, I. (2018). Matching Depth-Rotated Faces at Varying Degrees of Physical Similarity. Poster presented at the annual meeting of the Vision Sciences Society, St. Petersburg Beach, FL. May. doi: [10.1167/18.10.932](https://doi.org/10.1167/18.10.932).
- Biederman, I., Margalit, E., Maarek, R.S., Meschke, E.X., Shilowich, B.S., **Hacker, C. M.**, Juarez, J.J., Seamans, T. J. & Herald, S.B. (2017). What is the Nature of the Perceptual Deficit in Congenital Prosopagnosia? Poster presented at the annual meeting of the Vision Sciences Society, St. Petersburg Beach, FL. May. doi: [10.1167/17.10/619](https://doi.org/10.1167/17.10/619).

HONORS AND AWARDS

- Jameson-Hurvich Travel Award, Recipient** 2022, 2025
- NSF Graduate Research Fellowship, Honorable Mention** 2020
- [USC Discovery Scholar](#), Distinction recipient, Prize finalist** 2019
Graduation distinction awarded to students who excel in the classroom while demonstrating the ability to create exceptional new scholarship.
- [USC Neuroscience Outstanding Student of the Year Award](#), Recipient** 2019
Award given to USC's best neuroscience student with senior standing.
- [Brian Phillip Rakusin Neuroscience Scholarship Award](#), Recipient** 2018
\$10,000 Scholarship awarded each year to the most outstanding sophomore or junior demonstrating exceptional achievements and aspirations in the field of Neuroscience.
- [USC Provost's Undergraduate Research Fellowship](#), Six-time Recipient** 2017 – 2019
Fellowship awarded to select undergraduates demonstrating excellent academic standing and engaged in research, total value of \$8,000 over six semesters.

TEACHING

- Computer Vision and Imaging Module, University of Birmingham** 2024
Invited Guest Lecturer
Invited to give a virtual 45-minute lecture on the relationship between neuroscience and computer science for senior undergraduates.
- [Computational Neuroscience: Vision](#), Cold Spring Harbor Laboratory** 2024, 2026
Teaching Assistant
Course Organizers: Jonathan Pillow, Emily Cooper, John Serences, Lea Duncker
From a group of 24 students I was one of two selected to return as a TA the following year. In this two-week course I discussed course material and projects one-on-one with students, maintained the course website, and gave a lecture about my research.
- Computational and Theoretical Neuroscience, University of Pennsylvania** 2023
Recitation Leader
Taught a weekly "computational tutorial" reviewing mathematical concepts for undergraduate and graduate students with a neuroscience background.

TEACHING (CONT.)

CORE II: The Electrical Language of Cells, University of Pennsylvania **2020 – 2022**
Recitation Leader

Taught weekly recitations for first-year graduate students to supplement lectures on electrical and chemical signaling in excitable cells.

Cognitive Neuroscience, University of Pennsylvania **2021**
Teaching Assistant

Presented a lecture about object recognition, held office hours, and graded exams for an introductory undergraduate course.

OUTREACH AND SERVICE

[Penn NeuroKnow](#) **2021 – 2026**
Writer (2021-2026), Co-Editor (2022-2026)

Student-curated blog with posts about neuroscientific research written for the public. I have written 27 posts and edited over 125.

[GLIA](#) **2019 – 2026**
Member (2019-2026), Co-Director (2022-2023)

Coalition of neuroscience graduate students organizing outreach and professional development events. As co-director I oversaw the eight-member executive board and allocation of a \$36,500 budget.

Simons Collaboration on the Global Brain News Site **2022, 2025**
Writer

Contributed journalistic news pieces written for a broad neuroscience audience about [internal brain states](#) and the [International Brain Laboratory](#).

CNI +/- Seminar **2021 – 2023**
Organizer

Seminar for students and postdocs of the Computational Neuroscience Initiative to present and get feedback about ongoing research.